AIDCF-018-THM Cooling, Thermal and Fluid Infrastructure Domain Guide

Document control

Element	Value
Document ID	AIDCF-018-THM
Title	Cooling, Thermal and Fluid Infrastructure Domain Guide
Version	1.0
Date	2026-10-10
Status	Draft for controlled review
Author	Siva Sethumadhavan
Domain	THM (Cooling, Thermal and Fluid Infrastructure)
Controls covered	MCF-THM-001 to MCF-THM-024 (families THM-01 to THM-06)
Normative basis	AIDCF-007A (control registry), AIDCF-008 (evidence model), AIDCF-010 (scoring and gates)
Related documents	AIDCF-003, AIDCF-004, AIDCF-009, AIDCF-013, AIDCF-014, AIDCF-018-POW, AIDCF-018-CMP, AIDCF-018-ENV

Version	Date	Summary
1.0	2026-10-10	Initial issue: domain guidance aligned to the rewritten THM control records.

Requirements in this guide are identified DG-THM-REQ-NNN. The control records in AIDCF-007A remain the normative statement of each control; where this guide restates a control, the registry record prevails. Keywords follow AIDCF usage: **shall** is a requirement, **should** a recommendation, **may** a permission.

1 Purpose and scope

This guide explains how to apply the 24 THM controls when assessing whether a Capability Instance can continue to remove the heat generated by its AI infrastructure across the declared operating states. It is written for assessors, facility and thermal engineering authorities, and AI platform operators who need a shared view of what "thermal continuity" means for accelerator-dense facilities and what evidence demonstrates it.

The scope is the complete heat-removal path, from the heat source at the accelerator package to the final heat sink:

- **Air-side systems**: computer-room air handlers (CRAH), fan walls, in-row coolers, rear-door heat exchangers, hot- or cold-aisle containment and airflow management, including the residual air-side heat from liquid-cooled racks.
- **Liquid systems**: direct-to-chip cold plates, rack manifolds and quick-disconnect couplings, the technology cooling system (TCS, the secondary loop serving IT), coolant distribution units (CDUs, either liquid-to-liquid or liquid-to-air), immersion tanks and their heat exchangers.
- **Facility water and heat rejection**: the facility water system (FWS), chillers, dry coolers, adiabatic coolers, cooling towers and the condenser water system, thermal storage, make-up water and any heat-reuse or district-energy interface.

- **Fluids**: water, water-glycol solutions, dielectric immersion fluids and refrigerants, their chemistry, filtration and hazards.
- **Controls and instrumentation**: CDU controllers, BMS sequences, chiller plant managers, sensors and leak detection, as they affect heat removal.

Out of scope here (but linked through §14): electrical supply to mechanical plant (POW), environmental performance metrics such as PUE and WUE (ENV), fire detection and suppression (PHY), cyber security of controllers (SEC), and the accelerator's own thermal management and throttling behaviour (CMP).

2 Why this domain matters for AI capability

Accelerator-dense AI infrastructure changes the thermal problem in four ways that make conventional data-centre assumptions unsafe.

Density and liquid dependence. Racks for current training platforms commonly draw tens to well over a hundred kilowatts, most of which is captured by direct-to-chip liquid cooling. Air alone cannot remove this heat. The heat-removal path therefore includes pumps, heat exchangers, valves and fluid inventory that have no equivalent in an air-cooled hall, and each is a potential single point of failure.

Low thermal mass and short ride-through. A cold plate and its local coolant hold very little heat. When coolant flow stops, accelerator temperatures rise within seconds, and devices throttle or trip to protect themselves. The margin between "pumps stop" and "capability lost" is therefore a matter of seconds to minutes, not the tens of minutes an air-cooled room with large air volume might tolerate. Thermal ride-through during utility loss, UPS transfer, generator start and chiller restart is a first-order design and assurance concern (MCF-THM-003).

Synchronous load transients. Large training jobs run thousands of accelerators in lock-step. Compute phases and communication or checkpoint phases alternate, and job start, stop or failure moves the whole cluster's power and heat together. Cooling controls tuned for slowly varying enterprise loads can oscillate or lag when hundreds of kilowatts step simultaneously (MCF-THM-021).

Concentration of capability. A single CDU group or FWS branch may serve the scale-up domains on which an entire training run depends. A cooling fault that affects one fault domain can halt a job across the cluster, because synchronous training generally stops when any participating domain is lost. Thermal fault domains must therefore be understood in terms of the Service Outcomes they carry, not only the kilowatts they remove.

The consequence for assurance is that a thermal claim must cover the whole path, the transients and the degraded states. Steady-state capacity at mild ambient on a lightly loaded hall says little about continuity under full synchronous load on the hottest day during a power event. AIDCF-008 §49 states the same limit: no generalisation from steady low load to full transient AI demand.

3 Reference patterns and variants

AIDCF-013 provides the generic system cards for air and liquid cooling. The patterns below are the variants an assessor is most likely to meet; most facilities combine several.

Pattern	Description	Characteristic assurance concerns
P1 Air-cooled with containment	CRAH or fan walls supplying contained cold aisles or extracting from contained hot aisles; chilled water from chillers with or without economiser	Inlet temperature compliance at rack height; bypass and recirculation; CRAH redundancy at actual chilled-water temperature; containment–fire interface
P2 Rear-door heat exchanger	Passive or active water-cooled doors on each rack, fed from the FWS or a dedicated loop	Water in the IT row; door fan power; leak detection at each door; capacity at actual water temperature

Pattern	Description	Characteristic assurance concerns
P3 Direct-to-chip with liquid-to-liquid CDU	Cold plates on accelerators (and often CPUs, memory, NICs) on a TCS loop; CDUs exchange heat to the FWS	CDU pump redundancy; approach temperature; W-class compatibility; TCS integrity; quick-disconnect management; residual heat to air
P4 Direct-to-chip with liquid-to-air CDU	CDU rejects TCS heat to hall air through a coil; used where no FWS is available, typically in retrofits	Air-side capacity must absorb the full liquid load; hall temperature drives CDU capacity; air and liquid paths become one fault domain
P5 Single-phase immersion	IT submerged in dielectric fluid in tanks; fluid circulated to a heat exchanger on the FWS	Fluid compatibility with components; fluid degradation; handling hazards; tank leak containment; service procedures
P6 Chiller-less high-temperature water	FWS at W32 or above, rejected through dry coolers or adiabatic coolers without mechanical refrigeration	Capacity at extreme ambient; adiabatic water dependency and legionella control; no chiller restart delay but full ambient exposure
P7 Chilled-water with chillers and thermal storage	Mechanical refrigeration with chilled-water storage to bridge restart	Chiller restart-to-capacity; storage sizing for ride-through; refrigerant safety
P8 Heat reuse or district interface	Heat exported to a district network or adjacent consumer	Exported heat path must not be the only heat sink; fall-back rejection required

3.1 Technology cooling system versus facility water system

The distinction between the TCS and the FWS is central to direct-to-chip assurance. In the ASHRAE TC 9.9 liquid-cooling terminology, the **technology cooling system** is the loop that contacts IT equipment (cold plates, manifolds, couplings) and must meet the IT OEM's requirements for fluid, cleanliness, temperature, flow and pressure. The **facility water system** carries heat from CDUs to heat rejection and is designed to facility standards. The CDU is the boundary between them. Most IT-side failure modes (fouling, leaks at couplings, flow imbalance) arise in the TCS; most capacity and ride-through failure modes arise in the FWS and heat rejection. Assessors should therefore expect two sets of evidence, two owners (frequently the AI platform operator or OEM for the TCS and the facility operator for the FWS) and a documented interface contract between them (AIDCF-013 fluid and thermal interface contracts).

3.2 Liquid-cooling classes

ASHRAE TC 9.9 classifies facility water supplied to liquid-cooled IT by the maximum facility water supply temperature at the interface: classes W17, W27, W32, W40, W45 and W+ (above 45 °C). The class determines whether mechanical refrigeration is needed at the site's climate, how much approach temperature is available across the CDU, and whether heat reuse is practical. The design basis (MCF-THM-009) should state the class adopted and show that, at the worst-case facility water supply temperature for that class plus the CDU approach temperature, the TCS supply temperature remains within the accelerator OEM limit. Air-side environments are classified separately; the ASHRAE TC 9.9 recommended envelope for classes A1–A4 is 18–27 °C dry-bulb at the IT inlet, and later editions added a high-density air class with a narrower envelope. Site-specific OEM limits take precedence where they are tighter.

4 Required functions and capability linkage

The THM domain implements the following Required Functions (AIDCF-014 class RequiredFunction). Each is linked to Service Outcomes through the canonical chain (Capability Instance → Service Outcome → Required Function → InfrastructureSystem).

Required Function	Description	Principal controls
RF-THM-1 Capture heat	Transfer heat from accelerator packages and other components into air or coolant within OEM limits	MCF-THM-005, -006, -009, -012
RF-THM-2 Transport heat	Move coolant and air through TCS, CDUs and FWS at required flow and pressure	MCF-THM-007, -010, -011
RF-THM-3 Reject heat	Transfer heat to ambient or another sink at the design ambient	MCF-THM-004, -013, -014, -016
RF-THM-4 Maintain fluid fitness	Keep coolant and water chemically, physically and microbiologically within limits	MCF-THM-017, -018
RF-THM-5 Detect and contain leaks	Detect, locate, isolate and drain liquid releases within their fault domain	MCF-THM-019, -020
RF-THM-6 Control and sense	Keep temperatures and pressures stable under transient load and fail safe on controller or sensor loss	MCF-THM-021, -022
RF-THM-7 Sustain through states	Maintain heat removal through component loss, maintenance and power events	MCF-THM-002, -003, -015, -023
RF-THM-8 Recover	Restore heat removal and readmit IT load in a controlled sequence	MCF-THM-024

The linkage to capability is quantitative: the heat-load model (MCF-THM-001) converts the Capability Instance's compute deployment into heat per fault domain; capacity controls test that heat against what each stage can remove; and usable compute capacity (MCF-CMP-009) reports the accelerators that are throttled because the thermal function is not fully delivered. A thermal shortfall therefore appears in two places: as a THM control result and as a reduction in usable accelerator capacity. Assessors should reconcile the two.

5 Dependencies, fault domains, common-mode concerns

5.1 Dependencies

The heat-removal path depends on:

- **Electrical supply** to pumps, CDUs, fans, chillers and controls. The critical question is which of these are on UPS and which only on generator-backed supply (POW-03, POW-05).
- **Water supply** for evaporative and adiabatic heat rejection and for loop make-up (MCF-THM-015; ENV-03).
- **Controls networks and BMS** for sequencing and setpoints (MCF-THM-021; SEC-02).
- **Ambient conditions** at the heat-rejection plant, including local recirculation (MCF-THM-004, -014).
- **OEM requirements and support** for TCS fluid, cold-plate design and CDU firmware (DEP-03, SUP-04).
- **Fluid and chemical supply chains** for glycol, inhibitors, immersion fluid and refrigerant (SUP-05, ENV-05).
- **Third-party plant** in colocation or shared campuses (MCF-THM-016; DEP-05).

5.2 Fault domains

A thermal fault domain is the set of IT equipment that loses cooling together when a single element fails. Typical nesting is: cold plate → server tray → rack manifold → CDU (or CDU group) → TCS loop → FWS branch or header → chiller or dry-cooler train → central plant. Air-side fault domains (CRAH zone, containment pod) overlap liquid fault domains and do not usually align with them. The assessor should obtain or construct the fault-domain table that maps each rack to every thermal element it depends on (MCF-THM-001 requires this apportionment) and then compare it with the compute fault-domain mapping (MCF-CMP-008).

5.3 Common-mode concerns

Redundancy in the THM domain is frequently defeated by elements that are shared across nominally independent trains. Common modes the assessor should look for explicitly include:

- a single FWS header or a single pair of headers with a common isolation valve;
- one controller or PLC sequencing all CDUs or all chillers, with no independent fail-safe;
- CDU pumps for redundant groups fed from the same UPS output or distribution board;
- a shared make-up water supply, water-treatment dosing system or expansion vessel;
- a single BMS server or network switch on which all cooling setpoints depend;
- identical CDU firmware deployed to all units at once (a software common mode analogous to MCF-CMP-015);
- a coolant batch contaminated at source and used to fill all loops;
- heat-rejection plant co-located so that one recirculation plume or one roof fire affects all trains;
- a single external heat sink (district network) without independent fall-back.

Each identified common mode should be recorded with its treatment and owner (MCF-THM-016) and, where it affects the outcome, raised to CG-03 and CG-09 evaluation.

6 Threats, hazards and failure scenarios with propagation paths to Service Outcomes

AIDCF-004 defines the scenario record and the cross-domain library. The scenarios below are those most material for THM, written as propagation paths (initiating event → affected entities → safeguards → residual capability → outcome effect).

ID	Scenario (AIDCF-004 reference)	Propagation path	Outcome effect	Principal safeguards
THM-S1	Cooling loss at sustained AI load (X02)	Duty CDU pump fails → standby fails to start → TCS flow drops in one CDU group → accelerator temperatures rise in seconds → throttling then thermal trip → synchronous job halts across the cluster	Training throughput lost for the job; recovery time set by checkpoint interval and restart	MCF-THM-010 pump failover; MCF-THM-021 fail-safe; MCF-CMP-008 fault-domain spread
THM-S2	Utility loss with mechanical cooling interruption (X01 combined)	Utility lost → UPS carries IT and, if designed, CDU pumps → chillers or dry-cooler fans stop until generator → chillers take minutes to restart → TCS supply temperature rises → throttling	Throughput reduced or lost during transient; repeated trips if readmission is uncontrolled	MCF-THM-003 ride-through; MCF-THM-013 restart; MCF-THM-024 staged readmission
THM-S3	Liquid leak with detection gap (X03)	Coupling seal fails during rack service → drip onto power shelf → undetected because tray not covered → electrical fault → rack loss; release continues until manual isolation	Rack and possibly row lost; life-safety and electrical hazard	MCF-THM-012 coupling procedure; MCF-THM-019 detection; MCF-THM-020 isolation and drainage
THM-S4	Extreme ambient plus water restriction (X10)	Heatwave exceeds design dry-bulb → dry-cooler capacity falls → adiabatic trim needed → water restriction limits adiabatic use → FWS temperature rises → CDU approach consumed → TCS above limit	Forced load reduction across the hall	MCF-THM-004 derating; MCF-THM-015 water continuity; load-management procedure
THM-S5	Shared control failure (X05)	BMS or CDU controller fault drives valves closed or pumps to minimum → loss of flow on multiple loops simultaneously	Multi-fault-domain cooling loss defeating redundancy	MCF-THM-021 fail-safe states; MCF-THM-022 sensor plausibility; SEC-02

ID	Scenario (AIDCF-004 reference)	Propagation path	Outcome effect	Principal safeguards
THM-S6	Coolant quality degradation	Inhibitor depletion or biological growth → fouling of microchannel cold plates → rising ΔT and falling flow → localised throttling → hardware damage over time	Gradual capacity erosion; hidden throttling	MCF-THM-017, -018; MCF-CMP-017 throttle telemetry
THM-S7	Maintenance removes hidden redundancy (X04)	Planned CDU service while a second unit is degraded → group at N−1 below load → excursion on next transient	Unplanned capability loss during maintenance	MCF-THM-023; MCF-THM-003 maintenance-state matrix
THM-S8	Legionella release from evaporative plant	Inadequate treatment → bacterial growth in tower → aerosol drift → exposure of staff or public	Life-safety event; regulatory shutdown of evaporative cooling, causing capacity loss	MCF-THM-018 legionella scheme
THM-S9	Refrigerant release	Leak in chiller machinery room → asphyxiation or toxicity hazard → room evacuated → chiller plant unattended	Life-safety hazard; potential capacity loss	MCF-THM-013 refrigerant detection and ventilation
THM-S10	Thermal recovery failure	Restart after outage → pumps restarted before controls → pressure surge and air lock → repeated trips on readmission	Prolonged outage beyond recovery tolerance	MCF-THM-024

Threats with intent (for example manipulation of setpoints through the BMS) are treated under SEC-02 and CG-15. In THM their consequence is the same as THM-S5, which is why fail-safe behaviour (MCF-THM-021) is evaluated independently of the cause.

7 Safeguards

Safeguards are organised by the AIDCF-004 functions prevent, detect, contain, correct and recover.

- **Prevent**: capacity with margin at design conditions (MCF-THM-002, -004, -007, -010, -013); redundancy and ride-through (MCF-THM-003); airflow and containment discipline (MCF-THM-005, -006); compatible fluids and chemistry (MCF-THM-017, -018); controlled coupling procedures (MCF-THM-012); stable controls (MCF-THM-021); planned maintenance with stated residual redundancy (MCF-THM-023).
- **Detect**: inlet air monitoring (MCF-THM-008); TCS pressure, flow, ΔT and make-up monitoring (MCF-THM-011); leak detection with location (MCF-THM-019); sensor plausibility (MCF-THM-022); accelerator throttle telemetry as a lagging indicator (MCF-CMP-017).
- **Contain**: isolation valves, drip trays and drainage sized to the fault domain (MCF-THM-020); fault-domain spread of compute so that one thermal domain does not halt the outcome (MCF-CMP-008).
- **Correct**: automatic pump failover (MCF-THM-010); high-ambient and load-management procedures (MCF-THM-004); corrective water treatment (MCF-THM-018).
- **Recover**: thermal recovery sequence and staged readmission (MCF-THM-024), aligned with power restoration (POW-06) and compute re-qualification (MCF-CMP-023).

A safeguard is credited only for the states in which it is evidenced. For example, a CDU's N+1 pump redundancy credited in NORMAL is not automatically credited in MAINTENANCE when one pump is out for service.

8 Critical-gate linkage

THM controls map individually to gates under the AIDCF-010 rule: a CRITICAL mapping means the gate cannot PASS unless the control is CONFORMING at its required claim type; a SUPPORTING mapping informs the gate evaluation. The mapping for this domain is:

Gate	CRITICAL controls	SUPPORTING controls
CG-01	—	MCF-THM-001
CG-02	—	MCF-THM-002
CG-03	—	MCF-THM-004, MCF-THM-015, MCF-THM-016
CG-04	MCF-THM-018	MCF-THM-006, MCF-THM-012, MCF-THM-013, MCF-THM-017, MCF-THM-019, MCF-THM-020, MCF-THM-023
CG-05	—	MCF-THM-003
CG-06	MCF-THM-002, MCF-THM-003, MCF-THM-004, MCF-THM-010, MCF-THM-011, MCF-THM-013, MCF-THM-015	MCF-THM-001, MCF-THM-005, MCF-THM-006, MCF-THM-007, MCF-THM-008, MCF-THM-009, MCF-THM-012, MCF-THM-014, MCF-THM-016, MCF-THM-017, MCF-THM-018, MCF-THM-019, MCF-THM-020, MCF-THM-021, MCF-THM-022, MCF-THM-023, MCF-THM-024
CG-07	MCF-THM-021	MCF-THM-010, MCF-THM-022
CG-09	MCF-THM-020	MCF-THM-003, MCF-THM-011, MCF-THM-016, MCF-THM-019
CG-10	MCF-THM-024	—
CG-14	—	MCF-THM-012, MCF-THM-023
CG-15	—	MCF-THM-021

Notes on the mapping:

- **CG-06 Thermal continuity** has seven CRITICAL THM controls covering the minimum argument that the heat-removal path is sufficient across states: as-built capacity (MCF-THM-002), redundancy and ride-through (-003), capacity at design ambient (-004), CDU capacity and pump failover (-010), TCS integrity (-011), chiller capacity where chillers exist (-013) and water continuity where heat rejection depends on water (-015). Where a control is not applicable (for example MCF-THM-013 at a chiller-less site), the NOT_APPLICABLE result requires approved rationale and the gate is evaluated on the remaining CRITICAL controls.
- **CG-04 Life safety and hazardous energy** receives MCF-THM-018 as CRITICAL because legionella control in evaporative systems is a direct life-safety safeguard. Refrigerant safety (MCF-THM-013), the containment–fire-safety interface (-006), fluid hazards (-017), coupling spills (-012), leak detection (-019), isolation (-020) and safe isolation during maintenance (-023) are SUPPORTING.
- **CG-07 Critical control-system integrity** receives MCF-THM-021 as CRITICAL: cooling controllers that do not fail safe can defeat every redundant train simultaneously.
- **CG-09 Failure containment** receives MCF-THM-020 as CRITICAL: a leak that cannot be isolated within its fault domain propagates electrical and thermal consequences beyond tolerance.
- **CG-10 Recovery readiness** receives MCF-THM-024 as CRITICAL: cooling restart and staged readmission are prerequisites for compute and data recovery.

The CG-06 evidence floor in AIDCF-008 Part XIII is S5 with classes E1+E2+E3+E4. Accordingly the CRITICAL CG-06 controls carry profile tier CP-5 and sufficiency floor S5.

9 Capability-profile parameters

The following parameters are drawn from the THM control records. Values are **illustrative defaults** for profile authors and are not requirements; each Capability Profile shall set its own values with the stated basis (DG-THM-REQ-001). Where the basis cites an external source family, the value is attributed to that source; otherwise it is a profile choice.

Parameter	Unit	Illustrative default	Basis	Used by
Rack design heat load (liquid-cooled AI rack)	kW per rack	120 (illustrative)	Platform OEM thermal design data; site-specific	MCF-THM-001
Liquid-capture ratio (direct-to-chip platform)	% of rack heat to liquid	80 (illustrative)	OEM thermal data or measured; varies by platform generation	MCF-THM-001
Synchronous load step	% of fault-domain design load within 1 s	50 (illustrative)	Workload characterisation of synchronous training; to be measured	MCF-THM-001
Model-to-measured reconciliation tolerance	% of measured heat load	±10 (illustrative)	Capability profile	MCF-THM-001
Design capacity margin above design heat load	%	10 (illustrative)	Capability profile; design authority	MCF-THM-002
Commissioning load-bank test level	% of design heat load	100 (illustrative)	ASHRAE Guideline 0 practice; site commissioning plan	MCF-THM-002
Declared redundancy for CDU stage	configuration	N+1 (illustrative)	Capability profile	MCF-THM-003
Chiller restart-to-capacity time after power interruption	s	300 (illustrative)	Manufacturer rapid-restart data; verify by test	MCF-THM-003
Thermal ride-through required	s	600 (illustrative)	Derived: restart time plus margin; capability profile	MCF-THM-003
Maximum TCS supply temperature rise during transient	K	5 (illustrative)	OEM coolant inlet limit minus normal setpoint	MCF-THM-003
Design ambient dry-bulb	°C	site-specific (illustrative: 0.4% annual extreme)	ASHRAE climatic design conditions or national equivalent	MCF-THM-004
Design ambient wet-bulb (evaporative systems)	°C	site-specific	ASHRAE climatic design conditions	MCF-THM-004
Climate-change allowance on design ambient	K	2 (illustrative)	Capability profile; regional projections	MCF-THM-004
High-ambient action threshold	% of derated capacity	90 (illustrative)	Capability profile	MCF-THM-004

Parameter	Unit	Illustrative default	Basis	Used by
IT inlet temperature envelope	°C	18–27	ASHRAE TC 9.9 recommended envelope (classes A1–A4); site may adopt narrower OEM limits	MCF-THM-005
Air-handler ΔT lower bound (bypass indicator)	K	8 (illustrative)	Capability profile	MCF-THM-005
Hot-spot closure time	h	24 (illustrative)	Capability profile	MCF-THM-005
Maximum leakage area of containment	% of containment area	declared by design (illustrative 2)	Design authority	MCF-THM-006
Air-side redundancy per zone	configuration	N+1 (illustrative)	Capability profile	MCF-THM-007
Entering-water temperature for rating	°C	site design value	Design basis	MCF-THM-007
Inlet sensor density	sensors per row	≥ 3 (illustrative)	Capability profile	MCF-THM-008
Sensor accuracy	K	±0.5 (illustrative)	Capability profile; instrument specification	MCF-THM-008
Calibration interval	months	12 (illustrative)	Instrument manufacturer / site metrology policy	MCF-THM-008
Facility water class at CDU interface	ASHRAE W-class	W32 (illustrative)	ASHRAE TC 9.9 liquid-cooling classes W17–W45 and W+	MCF-THM-009
TCS supply temperature to cold plates	°C	OEM-specified (illustrative 30–40)	Platform OEM specification	MCF-THM-009
CDU heat-exchanger approach temperature	K	3 (illustrative)	CDU manufacturer data	MCF-THM-009
TCS filtration rating	µm	50 (illustrative; OEM may require finer)	Platform OEM specification; OCP cold-plate guidance	MCF-THM-009
CDU pump failover recovery time	s	10 (illustrative)	CDU manufacturer data; verify by test	MCF-THM-010
Minimum TCS flow per rack	l/min	OEM-specified	Platform OEM specification	MCF-THM-010
CDU group redundancy	configuration	N+1 (illustrative)	Capability profile	MCF-THM-010

Parameter	Unit	Illustrative default	Basis	Used by
Pressure test level	× maximum operating pressure	1.5 (illustrative)	Piping code or manufacturer specification	MCF-THM-011
Pressure test hold time	min	30 (illustrative)	Piping code or manufacturer specification	MCF-THM-011
Unexplained make-up volume investigation threshold	l per week per loop	5 (illustrative)	Capability profile	MCF-THM-011
Coupling inspection interval	months	12 (illustrative)	Coupling manufacturer guidance	MCF-THM-012
Rated connect/disconnect cycles	cycles	manufacturer-rated	Coupling manufacturer data	MCF-THM-012
Chiller restart-to-capacity	s	300 (illustrative)	Manufacturer rapid-restart data	MCF-THM-013
Chiller redundancy	configuration	N+1 (illustrative)	Capability profile	MCF-THM-013
Approach temperature degradation threshold	K above commissioned	2 (illustrative)	Capability profile	MCF-THM-014
Water autonomy at peak consumption	h	48 (illustrative)	Capability profile; utility risk assessment	MCF-THM-015
Peak water consumption	m^3/h	site-calculated	Design calculation at design wet-bulb and load	MCF-THM-015
Glycol concentration	% by volume	OEM-specified (illustrative 25)	Platform OEM coolant specification; OCP propylene-glycol guidance	MCF-THM-017
TCS sampling frequency	per month	1 (illustrative)	OEM/fluid supplier guidance	MCF-THM-018
Out-of-limit corrective-action time	days	7 (illustrative)	Capability profile	MCF-THM-018
Legionella risk assessment review interval	months	24 or on change (illustrative)	Applicable national regulation/ACOP	MCF-THM-018
Leak detection functional test interval	months	12 (illustrative)	Manufacturer guidance; capability profile	MCF-THM-019
Alarm-to-acknowledgement time	min	5 (illustrative)	Capability profile	MCF-THM-019

Parameter	Unit	Illustrative default	Basis	Used by
Maximum release volume per isolated section	l	design-specific (illustrative 50)	Design authority	MCF-THM-020
Valve exercise interval	months	12 (illustrative)	Manufacturer guidance	MCF-THM-020
Settling time after load step	s	120 (illustrative)	Controls design; capability profile	MCF-THM-021
Temperature control band	K	±1 (illustrative)	Controls design	MCF-THM-021
Control-critical sensor calibration interval	months	12 (illustrative)	Instrument manufacturer / metrology policy	MCF-THM-022
Redundant-sensor divergence alarm	K	1.5 (illustrative)	Capability profile	MCF-THM-022
Minimum residual redundancy during maintenance	configuration	N (illustrative)	Capability profile; risk acceptance	MCF-THM-023
Thermal recovery time objective	min	30 (illustrative)	Capability profile	MCF-THM-024
IT readmission step	% of fault-domain design load	25 (illustrative)	Capability profile; recovering capacity	MCF-THM-024

DG-THM-REQ-001 A capability profile that adopts THM controls shall state a value and a basis for every parameter listed above that applies to its patterns, or shall record the parameter as not applicable with rationale.

DG-THM-REQ-002 An assessor shall not treat an illustrative default from this table as an acceptance criterion unless the applicable capability profile has adopted it.

10 Evidence expectations by claim type

Each THM control declares one claim type and the matching minimum Assurance Level. The distribution in this domain is:

Claim type	Required AL	Controls	Typical profile tier and sufficiency floor
DESIGN	AL2	MCF-THM-001, MCF-THM-009, MCF-THM-016, MCF-THM-017	CP-1/S3
IMPLEMENTATION	AL3	MCF-THM-002, MCF-THM-006, MCF-THM-007	CP-2/S3, CP-5/S5
OPERATION	AL4	MCF-THM-004, MCF-THM-005, MCF-THM-008, MCF-THM-011, MCF-THM-012, MCF-THM-014, MCF-THM-018, MCF-THM-022, MCF-THM-023	CP-3/S4, CP-4/S4, CP-5/S5
EFFECTIVENESS	AL5	MCF-THM-010, MCF-THM-013, MCF-THM-019, MCF-THM-020, MCF-THM-021	CP-4/S4, CP-5/S4, CP-5/S5

Claim type	Required AL	Controls	Typical profile tier and sufficiency floor
RESILIENCE	AL5	MCF-THM-003, MCF-THM-015, MCF-THM-024	CP-5/S5

Practical expectations by claim type:

- **DESIGN (AL2)**: design basis documents, P&IDs, OEM requirement documents, dependency maps and calculations. Evidence is E1 with E7 analysis. A DESIGN result says nothing about the as-built state; MCF-THM-001, -009, -016 and -017 are DESIGN claims and support, but do not substitute for, the capacity and integrity controls.
- **IMPLEMENTATION (AL3)**: as-built reconciliation and commissioning. Evidence is E1 plus E2 configuration exports (BMS point lists, CDU configuration, unit schedules) and E3 inspection, with E4 commissioning records where capacity is claimed. MCF-THM-002 is an IMPLEMENTATION claim but is CG-06 CRITICAL and therefore requires E4 load-bank commissioning at S5.
- **OPERATION (AL4)**: telemetry and records over a representative period that includes the relevant seasons and states. MCF-THM-004 requires a period including the peak-ambient season; MCF-THM-018 requires twelve months of water-quality results. Telemetry is only as good as its sensors (MCF-THM-022) and its completeness; gaps shall be reported.
- **EFFECTIVENESS (AL5)**: tests under defined adverse conditions — pump failover, chiller restart, leak isolation, step-load control response and leak-detection wet tests. E4 test records shall include authorisation, initial state, method, expected result, observed result and anomalies (AIDCF-008 E4).
- **RESILIENCE (AL5)**: integrated or scenario evidence across states — black-building or utility-loss tests with time-stamped coolant traces (MCF-THM-003), water switchover (MCF-THM-015) and recovery exercises (MCF-THM-024).

Evidence quality for thermal claims is most often limited by three issues: telemetry from uncalibrated or poorly placed sensors (reliability, R), tests performed at partial load or mild ambient (coverage, C, and freshness against current load), and commissioning records that predate densification (freshness, F). The assessor should rate these dimensions explicitly rather than accept a single "tested" statement.

DG-THM-REQ-003 Evidence of capacity or ride-through obtained at a heat load lower than the current admitted load shall not support a CONFORMING result for the current load unless extrapolation is performed by a documented method and reviewed independently.

DG-THM-REQ-004 Thermal telemetry used as E3 evidence for an OPERATION, EFFECTIVENESS or RESILIENCE claim shall be accompanied by evidence of sensor calibration status and data completeness for the period.

11 Assessment guidance

11.1 Sequencing the assessment

1. **Fix the demand.** Start with MCF-THM-001. Without a current heat-load model apportioned to fault domains, every capacity control lacks an acceptance figure. Reconcile the model to the IT power budget and to compute deployment records (MCF-CMP-001, -003).
2. **Trace the path.** Using the P&ID, trace each fault domain from rack to final heat sink and list each stage. Build the stage-by-stage capacity table required by MCF-THM-002 and find the weakest stage. In direct-to-chip halls the weakest stage is often the CDU group, the FWS pumps or the dry coolers at design ambient, rather than the chillers.
3. **Test the states.** For each CRITICAL CG-06 control, check which operating states the evidence covers: NORMAL, MAINTENANCE (one unit out), DEGRADED (one unit failed), EMERGENCY (utility loss,

extreme ambient) and RECOVERY. Record uncovered states as U5 coverage-gap UNKNOWN Records.

4. **Look for common modes.** Walk the plant with the dependency map (MCF-THM-016) and verify shared headers, controllers, power boards and water supplies physically.
5. **Check fluids and leaks.** Review chemistry and microbiological results, coupling management and leak detection coverage against the room layout.
6. **Close with controls and recovery.** Review fail-safe evidence and recovery exercises last, because they depend on understanding the plant.

11.2 Sampling

Sampling follows AIDCF-009. For THM the population is usually the set of fault domains, not individual sensors. Assessors should include, at minimum, the fault domain carrying the highest heat load, the fault domain serving the scale-up domains of the largest training job, and any fault domain modified since the last commissioning. For sensor and leak-detection controls, sample by zone so that every wetted zone type (CDU skid, rack manifold, rear door, under-floor or overhead pipework, immersion tank) is represented.

11.3 Interpreting results

- A capacity control with commissioning evidence that predates densification is **UNKNOWN** for the current load, not CONFORMING.
- A ride-through analysis without a transient test is sufficient for a DESIGN view but leaves MCF-THM-003 **UNKNOWN** at its RESILIENCE claim.
- A margin below the profile's required margin, with capacity above load, is **PARTIAL**, which under the AIDCF-010 roll-up rule yields CG-06 FAIL if the control is CRITICAL. The assessor should record the margin shortfall as a Finding and identify the load limit that would restore conformity.
- Throttling events in accelerator telemetry (MCF-CMP-017) during the period are evidence that challenges a CONFORMING thermal claim and shall be reconciled before the claim is accepted.

DG-THM-REQ-005 Where accelerator telemetry shows thermal throttling in the assessment period, the assessor shall record the events as challenging evidence against the relevant THM controls and shall resolve or bound the conflict before concluding those controls are CONFORMING.

DG-THM-REQ-006 The assessor shall record for each CG-06 CRITICAL control the operating states covered by the evidence, and shall record each declared operating state not covered as an UNKNOWN Record with its severity.

12 Testing safety

Thermal tests are among the most hazardous and capability-affecting activities in an AI facility. Pump failover, chiller restart, utility-loss and isolation tests can trip production IT within seconds, and pressure tests, refrigerant work and evaporative-plant sampling present direct hazards to people.

- **Authority.** Hazardous or capability-affecting tests shall be performed only under a permit-to-work and method statement approved by the site authorising engineer, with the AI platform operator's agreement on the load at risk (DG-THM-REQ-007). Assessors observe, review and, where authorised, witness; they do not operate plant.
- **Preference for existing evidence.** Before requesting a new disruptive test, the assessor should seek evidence from previous commissioning, scheduled tests and real events with full telemetry.
- **Abort criteria.** Live thermal tests shall define abort criteria based on TCS supply temperature, IT inlet temperature or accelerator temperature, and a person with authority to abort shall be present (DG-THM-REQ-008).

- **Pressure systems.** Pressure tests of TCS or FWS sections require pressure-equipment competence, isolation from IT and exclusion zones.
- **Biological and chemical hazards.** Sampling of evaporative systems requires respiratory protection according to the legionella risk assessment; coolant and immersion fluid handling follows the safety data sheet.
- **Refrigerants.** Only certified refrigeration personnel work on refrigerant circuits; machinery-room entry follows the refrigerant detection and ventilation status.
- **Electrical proximity.** Wet tests of leak detection use minimal water away from energised equipment.

DG-THM-REQ-007 A THM test that can interrupt heat removal from production IT, release fluid or pressure, or expose people to refrigerant, chemical or biological hazards shall not be performed or requested by an assessor without a permit-to-work and method statement approved by the site authorising engineer and the agreement of the AI platform operator.

DG-THM-REQ-008 Every live thermal test shall define abort criteria and name the person authorised to abort before the test begins.

13 UNKNOWN hot-spots

The following are the places where THM assessments most often end in UNKNOWN. Each is listed with its usual type (AIDCF-008 §35) and severity (canonical US scale).

Hot-spot	Typical type	Typical severity	Why
Liquid-capture ratio and residual air heat for a new platform	U1 Missing	US3	OEM data not yet published or not shared; affects air-side and CDU sizing
Thermal ride-through at current load	U5 Coverage gap	US4	Transient tests rarely repeated after densification
Secondary-loop autonomy at peak load (AIDCF-008 §49 example)	U5 Coverage gap	US4	Pump failover tested only at low load
Peak-ambient performance in the first year of operation	U5 Coverage gap	US4	No summer yet observed at full load
CDU controller fail-safe behaviour (vendor-supplied logic)	U2 Inaccessible / U8 Method	US4	Vendor logic not disclosed; fail-safe not tested
Third-party or landlord plant in colocation	U2 Inaccessible / U10 Dependency	US4	Evidence held by provider; CG-03 interface
Make-up water consumption at design wet-bulb	U1 Missing	US4 where water-dependent	Not calculated; autonomy cannot be asserted
Sensor calibration of control-critical points	U6 Source unverified	US3	Calibration not tracked for OEM-supplied sensors
Microbiological results for adiabatic systems	U1 Missing	US4 (CG-04)	Adiabatic pads treated as "not a tower" and omitted from scheme

DG-THM-REQ-009 An UNKNOWN affecting a CG-06 or CG-10 CRITICAL THM control shall be recorded at severity US4 or higher and shall carry the gate consequence defined in AIDCF-010.

14 Interfaces with other domains

Domain	Interface	What THM needs	What THM provides
POW	Power to mechanical plant; ride-through	UPS coverage of pumps, CDUs and controls; generator start and transfer times; load-step behaviour	Mechanical load profile including restart inrush; chiller restart sequence
CMP	Rack-scale liquid interface; throttling	Rack heat, liquid-capture ratio, coolant requirements (MCF-CMP-001, -003); throttle telemetry (MCF-CMP-017)	TCS supply temperature and flow within OEM limits; readmission limits after recovery (MCF-THM-024 → MCF-CMP-023)
STO	Storage heat and placement	Storage system heat load and location	Cooling fault-domain information for checkpoint-tier placement (MCF-STO-002)
NET	Switch heat; liquid-cooled switches	Fabric equipment heat and cooling method	Cooling fault domains for fabric fault-domain design (MCF-NET-004)
ENV	Efficiency, water, refrigerants, climate	Water source and WUE constraints; refrigerant regulation; climate adaptation	Consumption data; refrigerant inventory; design ambient basis
PHY	Fire, water zones, structure	Fire detection and suppression in containment; water zone segregation; floor loading of tanks	Leak and drainage design; containment layout
OPS	Procedures and maintenance	Permit-to-work, method statements, calibration, alarm management	Residual-redundancy statements; maintenance plans
SEC	Controller security	Access control and segmentation of BMS and CDU controllers	Fail-safe definitions and control-critical point list
RES	Recovery and fault domains	Recovery sequencing across domains; integrated tests	Thermal recovery time objective; thermal fault-domain map
DEP / SUP	Utilities, OEMs, fluids	Water utility dependency; OEM and fluid supply	Dependency map entries for heat rejection

15 Informative references

Document-level references only; no clause-level content is reproduced.

- ASHRAE TC 9.9, *Thermal Guidelines for Data Processing Environments*.
- ASHRAE TC 9.9, *Liquid Cooling Guidelines for Datacom Equipment Centers*.
- ASHRAE Guideline 0, *The Commissioning Process*.
- ASHRAE Standard 90.4, *Energy Standard for Data Centers*.
- ASHRAE Standard 169 and ASHRAE Handbook — Fundamentals (climatic design conditions).
- ASHRAE Standard 188, *Legionellosis: Risk Management for Building Water Systems*; HSE ACOP L8 (UK).
- ASHRAE Standard 15, ISO 5149 and EN 378 (refrigeration system safety); EU F-gas Regulation or applicable refrigerant regulation.
- EN 50600-2-3 / ISO/IEC 22237-3 (data centre environmental control).
- ANSI/TIA-942; Uptime Institute Tier Standard: Topology.
- Open Compute Project Advanced Cooling Solutions guidance (cold plate, CDU and immersion).
- ASME B31.9 (building services piping); Pressure Equipment Directive 2014/68/EU or equivalent.
- IEC 62443 series; IEC 61131-3.
- NFPA 75 (fire protection of information technology equipment).
- Regulation (EC) No 1272/2008 (CLP) and GHS safety data sheets.
- ISO 22301 (business continuity management systems).
- AIDCF-004, AIDCF-007A, AIDCF-008, AIDCF-009, AIDCF-010, AIDCF-013, AIDCF-014.

16 Control summary

Control ID	Title	Type	AL	Claim type	Gate mappings
MCF-THM-001	Heat-Load Model	DIRECTIVE	AL2	DESIGN	CG-06 SUPPORTING; CG-01 SUPPORTING
MCF-THM-002	Design Capacity	PREVENTIVE	AL3	IMPLEMENTATION	CG-06 CRITICAL; CG-02 SUPPORTING
MCF-THM-003	Resilient Capacity	PREVENTIVE	AL5	RESILIENCE	CG-06 CRITICAL; CG-09 SUPPORTING; CG-05 SUPPORTING
MCF-THM-004	Seasonal Derating	PREVENTIVE	AL4	OPERATION	CG-06 CRITICAL; CG-03 SUPPORTING
MCF-THM-005	Airflow Management	PREVENTIVE	AL4	OPERATION	CG-06 SUPPORTING
MCF-THM-006	Containment	PREVENTIVE	AL3	IMPLEMENTATION	CG-06 SUPPORTING; CG-04 SUPPORTING
MCF-THM-007	CRAH and In-Row Capacity	PREVENTIVE	AL3	IMPLEMENTATION	CG-06 SUPPORTING
MCF-THM-008	Air-Side Monitoring	DETECTIVE	AL4	OPERATION	CG-06 SUPPORTING
MCF-THM-009	Direct-to-Chip Architecture	DIRECTIVE	AL2	DESIGN	CG-06 SUPPORTING
MCF-THM-010	CDU Capacity	PREVENTIVE	AL5	EFFECTIVENESS	CG-06 CRITICAL; CG-07 SUPPORTING
MCF-THM-011	Secondary Loop Integrity	PREVENTIVE	AL4	OPERATION	CG-06 CRITICAL; CG-09 SUPPORTING
MCF-THM-012	Quick-Disconnect Management	PREVENTIVE	AL4	OPERATION	CG-06 SUPPORTING; CG-04 SUPPORTING; CG-14 SUPPORTING
MCF-THM-013	Chiller Capacity	PREVENTIVE	AL5	EFFECTIVENESS	CG-06 CRITICAL; CG-04 SUPPORTING
MCF-THM-014	Cooling Tower or Dry Cooler	PREVENTIVE	AL4	OPERATION	CG-06 SUPPORTING
MCF-THM-015	Water Continuity	PREVENTIVE	AL5	RESILIENCE	CG-06 CRITICAL; CG-03 SUPPORTING
MCF-THM-016	Heat-Rejection Dependency	DIRECTIVE	AL2	DESIGN	CG-03 SUPPORTING; CG-09 SUPPORTING; CG-06 SUPPORTING
MCF-THM-017	Coolant Compatibility	PREVENTIVE	AL2	DESIGN	CG-04 SUPPORTING; CG-06 SUPPORTING
MCF-THM-018	Water Quality	PREVENTIVE	AL4	OPERATION	CG-04 CRITICAL; CG-06 SUPPORTING
MCF-THM-019	Leak Detection	DETECTIVE	AL5	EFFECTIVENESS	CG-06 SUPPORTING; CG-09 SUPPORTING; CG-04 SUPPORTING
MCF-THM-020	Isolation and Drainage	CORRECTIVE	AL5	EFFECTIVENESS	CG-09 CRITICAL; CG-04 SUPPORTING; CG-06 SUPPORTING
MCF-THM-021	Control Stability	PREVENTIVE	AL5	EFFECTIVENESS	CG-07 CRITICAL; CG-06 SUPPORTING; CG-15 SUPPORTING
MCF-THM-022	Sensor Trust	DETECTIVE	AL4	OPERATION	CG-07 SUPPORTING; CG-06 SUPPORTING

Control ID	Title	Type	AL	Claim type	Gate mappings
MCF-THM-023	Maintenance States	PREVENTIVE	AL4	OPERATION	CG-06 SUPPORTING; CG-14 SUPPORTING; CG-04 SUPPORTING
MCF-THM-024	Thermal Recovery	RECOVERY	AL5	RESILIENCE	CG-10 CRITICAL; CG-06 SUPPORTING

17 Worked synthetic example

Synthetic example. All names, figures and results below are invented for illustration and do not describe any real facility, design or assessment.

Context. Capability Instance CI-SYN-THM-01 is a 2,048-accelerator training cluster in Hall B of a synthetic facility. The Service Outcome is sustained training throughput with interruption not exceeding the job's checkpoint-restart tolerance. The cluster comprises 32 racks of 64 accelerators. The capability profile adopts a rack design heat load of 120 kW, a liquid-capture ratio of 80 %, a design capacity margin of 10 % and N+1 CDU redundancy. Pattern P3 (direct-to-chip, liquid-to-liquid CDUs) is combined with P6 (W32 facility water, dry coolers with adiabatic trim, no chillers).

Demand (MCF-THM-001). Total IT heat is 32 × 120 kW = 3,840 kW, of which 3,072 kW goes to liquid and 768 kW to air. Racks are served by four CDU groups of eight racks, so each group carries 8 × 120 kW × 0.8 = 768 kW of liquid heat. The model, however, takes the 80 % capture ratio from OEM data stated at a TCS supply temperature lower than the site's design value, and the OEM data give a lower ratio at the site's temperature; the model owner had not updated the figure. The assessor records a Finding and a U1 UNKNOWN for the residual air heat, US3, pending OEM confirmation.

Capacity (MCF-THM-002, -010). Each CDU group has three CDUs rated 450 kW at the design FWS supply temperature; with N+1, demonstrated group capacity is 2 × 450 kW = 900 kW against 768 kW, a margin of 17 %, which meets the 10 % profile margin. Commissioning load-bank records exist for all four groups. Pump failover tests were performed at commissioning when only 16 racks were installed; the current load is double. MCF-THM-010 (EFFECTIVENESS) is therefore **UNKNOWN** for current load (U5, US4).

Ride-through (MCF-THM-003). CDU pumps and controls are on UPS; dry-cooler fans are on generator-backed supply only. The ride-through calculation uses a combined TCS and FWS inventory of about 12 m³ of water-glycol (heat capacity roughly 47 MJ/K): at 3,072 kW of liquid heat the bulk coolant temperature rises at about 0.065 K/s while heat rejection is lost, reaching the 5 K transient limit in roughly 75 s. Generator start and fan restart are designed to complete within 30 s. The calculation is credible, but no integrated utility-loss test has been performed since densification. MCF-THM-003 is **UNKNOWN** (U5, US4).

Ambient (MCF-THM-004). The design ambient basis is documented from a named climatic dataset with a 2 K climate allowance. The facility has operated for eight months, none in the peak season. MCF-THM-004 is **UNKNOWN** (U5, US4) for EMERGENCY and peak-season states.

Water (MCF-THM-015, -018). Adiabatic trim depends on mains water with 24 h of storage against a profile requirement of 48 h. Capacity is above load but autonomy is below requirement: MCF-THM-015 is **PARTIAL**. The legionella risk assessment covers the adiabatic pads with monthly microbiological sampling, all within limits: MCF-THM-018 is **CONFORMING**.

Not applicable. MCF-THM-013 (Chiller Capacity) is recorded NOT_APPLICABLE with approved rationale: the site has no mechanical refrigeration, and dry-cooler capacity at design ambient is covered by MCF-THM-004.

Gate Result. Under the AIDCF-010 roll-up rule, CG-06 is **FAIL** because a CRITICAL control (MCF-THM-015) is PARTIAL. Had MCF-THM-015 been CONFORMING, CG-06 would still have been UNKNOWN because MCF-

THM-003, -004 and -010 are UNKNOWN. CG-04 has no failing THM input from this example. The decision consequence is determined by the CG-06 ceiling in AIDCF-010.

Actions recorded. (1) Increase water storage or agree a tested tanker supply to reach 48 h autonomy, or adopt a profile tolerance with approved load reduction in water-restricted states; (2) repeat CDU pump failover tests at current load in a planned window; (3) perform an integrated utility-loss test with time-stamped coolant traces, or a calibrated simulation reviewed independently; (4) obtain OEM liquid-capture data at site TCS temperature and update the heat-load model; (5) reassess MCF-THM-004 after the first peak season. Each action carries a reassessment trigger for the affected controls.